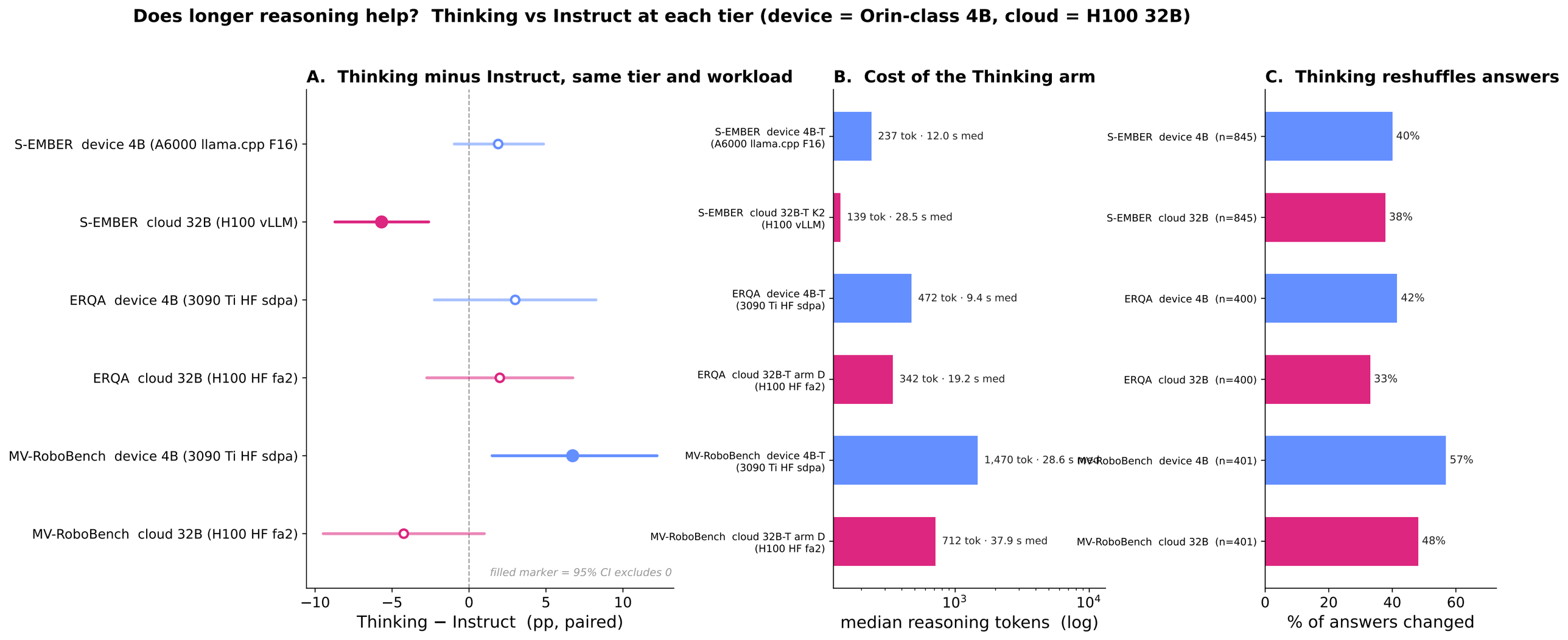
Does longer reasoning help? Thinking vs Instruct at each tier (device = Orin-class 4B, cloud = H100 32B)
A. Thinking minus Instruct, same tier and workload
S-EMBER device 4B (A6000 llama.cpp F16)
S-EMBER cloud 32B (H100 vLLM)
ERQA device 4B (3090 Ti HF sdpa)
ERQA cloud 32B (H100 HF fa2)
MV-RoboBench device 4B (3090 Ti HF sdpa)
MV-RoboBench cloud 32B (H100 HF fa2)
filled marker = 95% CI excludes 0
−10
−5
0
5
10
Thinking − Instruct (pp, paired)
B. Cost of the Thinking arm
S-EMBER device 4B-T (A6000 llama.cpp F16)
237 tok · 12.0 s med
S-EMBER cloud 32B-T K2 (H100 vLLM)
139 tok · 28.5 s med
ERQA device 4B-T (3090 Ti HF sdpa)
472 tok · 9.4 s med
ERQA cloud 32B-T arm D (H100 HF fa2)
342 tok · 19.2 s med
MV-RoboBench device 4B-T (3090 Ti HF sdpa)
1,470 tok · 28.6 s med
MV-RoboBench cloud 32B-T arm D (H100 HF fa2)
712 tok · 37.9 s med
10^3
10^4
median reasoning tokens (log)
C. Thinking reshuffles answers
S-EMBER device 4B (n=845)
40%
S-EMBER cloud 32B (n=845)
38%
ERQA device 4B (n=400)
42%
ERQA cloud 32B (n=400)
33%
MV-RoboBench device 4B (n=401)
57%
MV-RoboBench cloud 32B (n=401)
48%
0
20
40
60
% of answers changed